

B3: Atomic & Laser Physics — Model Solutions, Trinity 2024

Question 1 — Spin-orbit interaction and fine structure

Problem. One-electron atom: derive H_{SO} (Thomas factor via $g_s \rightarrow g_s - 1$), show the $l \neq 0$ splitting $\Delta E = Z^4 \alpha^2 \hbar c R_\infty / [n^3 l(l+1)]$, then apply to H ($2 \rightarrow 3$) vs. He⁺ ($4 \rightarrow 6$): fine structure, mass shift, E1 selection rules, Doppler width, and the Z giving fine structure equal to the $2 \rightarrow 3$ energy of H.

(a) Origin and derivation of H_{SO}

In the electron rest frame the orbiting nucleus produces a magnetic field that couples to the electron spin moment $\boldsymbol{\mu}_s = -g_s \mu_B \mathbf{S} / \hbar$.

Lorentz boost of the nuclear $\mathbf{E} = -\nabla\phi$ to the electron frame, $\mathcal{O}(v^2/c^2)$:

$$\mathbf{B}' = -\frac{1}{c^2} \mathbf{v} \times \mathbf{E}.$$

Central potential, $\mathbf{E} = -\hat{\mathbf{r}} \partial_r \phi$:

$$\mathbf{v} \times \mathbf{E} = -\partial_r \phi (\mathbf{v} \times \hat{\mathbf{r}}).$$

Use $\hat{\mathbf{r}} = \mathbf{r}/r$:

$$\mathbf{v} \times \mathbf{E} = -\frac{1}{r} \partial_r \phi (\mathbf{v} \times \mathbf{r}).$$

Identify $\mathbf{L} = m_e \mathbf{r} \times \mathbf{v} = -m_e \mathbf{v} \times \mathbf{r}$:

$$\mathbf{v} \times \mathbf{r} = -\frac{\mathbf{L}}{m_e}.$$

Insert in $\mathbf{B}'$:

$$\mathbf{B}' = -\frac{1}{m_e c^2} \frac{1}{r} \partial_r \phi \mathbf{L}.$$

Zeeman energy of spin moment in $\mathbf{B}'$:

$$H' = -\boldsymbol{\mu}_s \cdot \mathbf{B}' = \frac{g_s \mu_B}{\hbar} \mathbf{S} \cdot \mathbf{B}'.$$

Substitute $\mu_B = e\hbar/(2m_e)$:

$$H' = \frac{g_s e}{2m_e} \mathbf{S} \cdot \mathbf{B}'.$$

Substitute $\mathbf{B}'$:

$$H' = -\frac{g_s e}{2m_e^2 c^2} \frac{1}{r} \partial_r \phi \mathbf{L} \cdot \mathbf{S}.$$

Apply Thomas correction $g_s \rightarrow g_s - 1 \approx 1$ and write $U = -e\phi$ so $-e \partial_r \phi = \partial_r U$:

$$\boxed{H_{\text{SO}} = \frac{1}{2m_e^2 c^2} \frac{1}{r} \frac{\partial U}{\partial r} \mathbf{L} \cdot \mathbf{S}.} \quad (1)$$

Hydrogenic $U = -Ze^2/(4\pi\epsilon_0 r)$, so $\partial_r U = Ze^2/(4\pi\epsilon_0 r^2)$:

$$H_{\text{SO}} = \frac{Ze^2}{8\pi\epsilon_0 m_e^2 c^2 r^3} \mathbf{L} \cdot \mathbf{S} \equiv \xi(r) \mathbf{L} \cdot \mathbf{S}. \quad (2)$$

(b) Splitting of a configuration into two levels

Coupled basis $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2).$$

Take eigenvalues with $s = 1/2$, $l \neq 0$, $j = l \pm 1/2$:

$$\langle \mathbf{L} \cdot \mathbf{S} \rangle_j = \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)].$$

Evaluate $j = l + 1/2$, $j(j+1) = l^2 + 2l + 3/4$:

$$\langle \mathbf{L} \cdot \mathbf{S} \rangle_{j=l+1/2} = \frac{\hbar^2}{2} l.$$

Evaluate $j = l - 1/2$, $j(j+1) = l^2 - 1/4$:

$$\langle \mathbf{L} \cdot \mathbf{S} \rangle_{j=l-1/2} = -\frac{\hbar^2}{2} (l+1).$$

Take expectation value $\Delta E_j = \langle \xi(r) \rangle \langle \mathbf{L} \cdot \mathbf{S} \rangle_j$ and subtract:

$$\Delta E = \langle \xi(r) \rangle \left[\frac{\hbar^2}{2} l + \frac{\hbar^2}{2} (l+1) \right].$$

Collect:

$$\Delta E = \langle \xi(r) \rangle \frac{\hbar^2}{2} (2l+1).$$

Insert $\langle \xi(r) \rangle = \frac{Ze^2}{8\pi\epsilon_0 m_e^2 c^2} \langle 1/r^3 \rangle$ with hydrogenic $\langle 1/r^3 \rangle = 2(Z/na_0)^3/[l(l+1)(2l+1)]$:

$$\Delta E = \frac{Ze^2}{8\pi\epsilon_0 m_e^2 c^2} \cdot \frac{2Z^3}{n^3 a_0^3 l(l+1)(2l+1)} \cdot \frac{\hbar^2 (2l+1)}{2}.$$

Cancel the $(2l+1)$ factors:

$$\Delta E = \frac{Z^4 e^2 \hbar^2}{8\pi\epsilon_0 m_e^2 c^2 n^3 a_0^3 l(l+1)}.$$

Apply $e^2/(4\pi\epsilon_0) = \alpha\hbar c$ so $e^2/(8\pi\epsilon_0) = \alpha\hbar c/2$:

$$\Delta E = \frac{Z^4 \alpha \hbar^3 c}{2m_e^2 c^2 n^3 a_0^3 l(l+1)}.$$

Apply $a_0 = \hbar/(\alpha m_e c)$, so $1/a_0^3 = \alpha^3 m_e^3 c^3/\hbar^3$:

$$\Delta E = \frac{Z^4 \alpha^4 m_e c^2}{2n^3 l(l+1)}.$$

Apply $\hbar c R_\infty = \frac{1}{2} m_e c^2 \alpha^2$, so $\alpha^4 m_e c^2 = 2\alpha^2 \hbar c R_\infty$:

$$\Delta E = \frac{Z^4 \alpha^2 \hbar c R_\infty}{n^3 l(l+1)} \quad (l \neq 0).$$

(3)

(c) **H** ($2 \rightarrow 3$) **vs.** **He⁺** ($4 \rightarrow 6$)

Bohr energy $E_{n \rightarrow n'} = hcR_\infty Z^2 (1/n^2 - 1/n'^2)$.

H ($Z = 1, 2 \rightarrow 3$):

$$\Delta E^{\text{H}} = hcR_\infty \left(\frac{1}{4} - \frac{1}{9} \right) = \frac{9-4}{36} hcR_\infty = \frac{5}{36} hcR_\infty.$$

He⁺ ($Z = 2, 4 \rightarrow 6$):

$$\Delta E^{\text{He}^+} = 4hcR_\infty \left(\frac{1}{16} - \frac{1}{36} \right) = 4hcR_\infty \frac{36-16}{16 \cdot 36} = \frac{5}{36} hcR_\infty.$$

Same Bohr wavelength $\lambda \simeq 656 \text{ nm}$ (H_α), $\nu_0 \simeq 4.57 \times 10^{14} \text{ Hz}$.

Largest fine-structure splitting is at smallest $l \neq 0$, i.e. $l = 1$. Apply (3) for H, $Z = 1, n = 2, l = 1$:

$$\Delta E_{n=2, l=1}^{\text{H}} = \frac{1^4 \cdot \alpha^2}{2^3 \cdot 1 \cdot 2} hcR_\infty.$$

Reduce:

$$\Delta E_{n=2, l=1}^{\text{H}} = \frac{\alpha^2}{16} hcR_\infty.$$

Apply (3) for He⁺, $Z = 2, n = 4, l = 1$:

$$\Delta E_{n=4, l=1}^{\text{He}^+} = \frac{2^4 \alpha^2}{4^3 \cdot 1 \cdot 2} hcR_\infty = \frac{16}{128} \alpha^2 hcR_\infty.$$

Reduce:

$$\Delta E_{n=4, l=1}^{\text{He}^+} = \frac{\alpha^2}{8} hcR_\infty.$$

Divide H splitting by $h\nu_0 = \frac{5}{36} hcR_\infty$:

$$\frac{\Delta \nu^{\text{H}}}{\nu_0} = \frac{\alpha^2/16}{5/36} = \frac{36\alpha^2}{16 \cdot 5} = \frac{36\alpha^2}{80}.$$

Reduce:

$$\frac{\Delta \nu^{\text{H}}}{\nu_0} = \frac{9\alpha^2}{20}.$$

Insert $\alpha^2 \simeq 5.33 \times 10^{-5}$:

$$\frac{\Delta \nu^{\text{H}}}{\nu_0} \simeq \frac{9 \cdot 5.33 \times 10^{-5}}{20} \simeq 2.4 \times 10^{-5}.$$

Divide He⁺ splitting similarly:

$$\frac{\Delta \nu^{\text{He}^+}}{\nu_0} = \frac{\alpha^2/8}{5/36} = \frac{36\alpha^2}{40} = \frac{9\alpha^2}{10}.$$

Insert α^2 :

$$\frac{\Delta \nu^{\text{He}^+}}{\nu_0} \simeq \frac{9 \cdot 5.33 \times 10^{-5}}{10} \simeq 4.8 \times 10^{-5}.$$

Collect results:

$\frac{\Delta \nu^{\text{H}}}{\nu_0} = \frac{9\alpha^2}{20} \simeq 2.4 \times 10^{-5}, \quad \frac{\Delta \nu^{\text{He}^+}}{\nu_0} = \frac{9\alpha^2}{10} \simeq 4.8 \times 10^{-5}.$	(4)
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Mass shift. Reduced-mass Rydberg $R_M = R_\infty/(1 + m_e/M)$ shifts the transition frequency:

$$\frac{\nu(M)}{\nu_0} = \frac{R_M}{R_\infty} = \frac{1}{1 + m_e/M}.$$

Expand to leading order in m_e/M :

$$\frac{\nu(M)}{\nu_0} \simeq 1 - \frac{m_e}{M}.$$

Take the difference between H ($M = m_p$) and He⁺ ($M = M_\alpha \simeq 4m_p$):

$$\frac{\Delta\nu_{\text{mass}}}{\nu_0} = \frac{\nu(m_p) - \nu(4m_p)}{\nu_0} \simeq \frac{m_e}{m_p} - \frac{m_e}{4m_p}.$$

Collect:

$$\frac{\Delta\nu_{\text{mass}}}{\nu_0} \simeq \frac{3}{4} \cdot \frac{m_e}{m_p}.$$

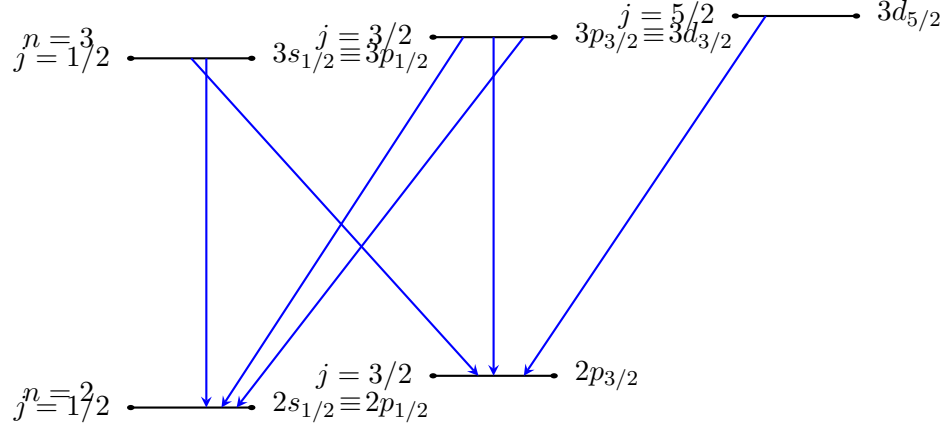
Insert $m_e/m_p \simeq 1/1836$:

$$\frac{\Delta\nu_{\text{mass}}}{\nu_0} \simeq \frac{3}{4} \cdot \frac{1}{1836} \simeq 4 \times 10^{-4},$$

an order of magnitude larger than the fine-structure splittings; mass shift dominates the wavelength difference.

(d) Energy-level diagram and E1 selection rules, H $n = 2 \rightarrow 3$

Levels $n^{2S+1}L_J$, $S = 1/2$. Dirac-degeneracy gives $2s_{1/2} \equiv 2p_{1/2}$, $3s_{1/2} \equiv 3p_{1/2}$, $3p_{3/2} \equiv 3d_{3/2}$.



E1 selection rules:

$$\Delta l = \pm 1, \quad \Delta s = 0, \quad \Delta j = 0, \pm 1 \ (0 \not\rightarrow 0), \quad \Delta m_j = 0, \pm 1. \quad (5)$$

Justification of $\Delta l = \pm 1$. $\hat{\mathbf{d}} = -e\hat{\mathbf{r}}$ is a rank-1 odd-parity tensor. Parity selection forces $l - l'$ odd; Wigner-Eckart forces $|l - l'| \leq 1$. Together: $\Delta l = \pm 1$.

(e) Doppler broadening at $T = 300$ K

Thermal Doppler FWHM:

$$\frac{\Delta\nu_D}{\nu_0} = \sqrt{\frac{8k_B T \ln 2}{Mc^2}}.$$

Insert $k_B T \simeq 4.14 \times 10^{-21}$ J, $M = m_p$, $c^2 \simeq 9 \times 10^{16} \text{ m}^2\text{s}^{-2}$:

$$\frac{\Delta\nu_D^H}{\nu_0} = \sqrt{\frac{8 \cdot 4.14 \times 10^{-21} \cdot 0.693}{1.67 \times 10^{-27} \cdot 9 \times 10^{16}}}.$$

Evaluate the radicand:

$$\frac{\Delta\nu_D^H}{\nu_0} = \sqrt{\frac{2.296 \times 10^{-20}}{1.503 \times 10^{-10}}} = \sqrt{1.527 \times 10^{-10}}.$$

Take the square root:

$$\frac{\Delta\nu_D^H}{\nu_0} \simeq 1.2 \times 10^{-5}.$$

He^+ has $M \simeq 4m_p$, so width scales by $1/\sqrt{M/m_p} = 1/\sqrt{4} = 1/2$:

$$\frac{\Delta\nu_D^{\text{He}^+}}{\nu_0} = \frac{1}{2} \cdot 1.2 \times 10^{-5}.$$

Collect both results:

$$\boxed{\frac{\Delta\nu_D^H}{\nu_0} \simeq 1.2 \times 10^{-5}, \quad \frac{\Delta\nu_D^{\text{He}^+}}{\nu_0} \simeq 6 \times 10^{-6}.} \quad (6)$$

Further mechanism in a discharge lamp. Pressure (collisional) broadening: random collisional dephasing of the radiating dipole, Lorentzian width $\sim 1/\tau_{\text{coll}}$. (Stark broadening from plasma microfields is also strong for H because of its linear Stark effect.)

(f) Fine structure equal to the H $2 \rightarrow 3$ energy

Equate $n = 2, l = 1$ splitting to $\frac{5}{36}hcR_\infty$:

$$\frac{Z^4 \alpha^2}{16} hcR_\infty = \frac{5}{36} hcR_\infty.$$

Cancel hcR_∞ and rearrange for Z^4 :

$$Z^4 = \frac{5 \cdot 16}{36 \alpha^2} = \frac{80}{36 \alpha^2} = \frac{20}{9 \alpha^2}.$$

Insert $\alpha^2 \simeq 5.33 \times 10^{-5}$:

$$Z^4 \simeq \frac{20}{9 \cdot 5.33 \times 10^{-5}} = \frac{20}{4.80 \times 10^{-4}} \simeq 4.17 \times 10^4.$$

Take square root:

$$Z^2 \simeq \sqrt{4.17 \times 10^4} \simeq 204.$$

Take square root again:

$$\boxed{Z \simeq \sqrt{204} \simeq 14.3.} \quad (7)$$

Alkali Z-scaling. Valence np electron penetrates the unscreened core; SO coupling samples the bare nuclear field. Empirical ΔE_{FS} : Na 0.42 meV, K 2.0 meV, Rb 6.96 meV, Cs 28 meV — super-quartic in Z because both the inner field and the penetrating-wavefunction amplitude grow with Z .

Question 2 — Hyperfine structure and the hydrogen MASER

Problem. Hamiltonian $\mathcal{H} = A \mathbf{I} \cdot \mathbf{J} + g_J \mu_B \mathbf{J} \cdot \mathbf{B} - g_I \mu_N \mathbf{I} \cdot \mathbf{B}$. Identify each term, find the weak/strong-field crossover for ground-state H ($A/h = 1420$ MHz, $I = 1/2$), derive the eigenenergies in both regimes, and analyse a Stern–Gerlach beam with $B_z = 1$ T, $\partial_z B_z = 20$ T/m as the basis of an H maser.

(a) Origin of each term

$A \mathbf{I} \cdot \mathbf{J}$: hyperfine coupling of nuclear moment $\boldsymbol{\mu}_I = g_I \mu_N \mathbf{I}/\hbar$ to the electron-generated magnetic field at the nucleus (dominated by the Fermi-contact term for s states).

$g_J \mu_B \mathbf{J} \cdot \mathbf{B}$: electronic Zeeman, energy of $\boldsymbol{\mu}_J = -g_J \mu_B \mathbf{J}/\hbar$ in $\mathbf{B}$.

$-g_I \mu_N \mathbf{I} \cdot \mathbf{B}$: nuclear Zeeman, smaller by $m_e/m_p \sim 1/1836$.

Regimes. Weak field $g_J \mu_B B \ll A$: $\mathbf{I}, \mathbf{J}$ couple to $\mathbf{F} = \mathbf{I} + \mathbf{J}$; good QNs $|F, m_F\rangle$. Strong field $g_J \mu_B B \gg A$: $\mathbf{I}, \mathbf{J}$ decouple; good QNs $|m_J, m_I\rangle$ (Paschen–Back of hyperfine).

(b) Cross-over field

Set the Zeeman shift equal to the hyperfine splitting $g_J \mu_B B_c \simeq A$.

Compute A from $\nu_{\text{HF}} = 1420$ MHz:

$$A = h \nu_{\text{HF}} = 6.63 \times 10^{-34} \text{ J s} \cdot 1.42 \times 10^9 \text{ Hz} \simeq 9.41 \times 10^{-25} \text{ J}.$$

With $g_J \simeq 2$ for the $1^2S_{1/2}$ ground state, solve for B_c :

$$B_c = \frac{A}{g_J \mu_B} = \frac{9.41 \times 10^{-25} \text{ J}}{2 \cdot 9.27 \times 10^{-24} \text{ J/T}}.$$

Evaluate:

$$\boxed{B_c \simeq 50 \text{ mT } (\sim 500 \text{ G}).} \quad (8)$$

(c) Weak-field eigenenergies

Take $\mathbf{I}, \mathbf{J}$ dimensionless (in units of $\hbar$). Define $\mathbf{F} = \mathbf{I} + \mathbf{J}$ and square:

$$\mathbf{F}^2 = \mathbf{I}^2 + \mathbf{J}^2 + 2\mathbf{I} \cdot \mathbf{J}.$$

Solve for $\mathbf{I} \cdot \mathbf{J}$:

$$\mathbf{I} \cdot \mathbf{J} = \frac{1}{2}[F(F+1) - I(I+1) - J(J+1)].$$

Insert $I = J = 1/2$, $F = 1$:

$$\langle \mathbf{I} \cdot \mathbf{J} \rangle_{F=1} = \frac{1}{2}[2 - \frac{3}{4} - \frac{3}{4}] = \frac{1}{4}.$$

Insert $F = 0$:

$$\langle \mathbf{I} \cdot \mathbf{J} \rangle_{F=0} = \frac{1}{2}[0 - \frac{3}{4} - \frac{3}{4}] = -\frac{3}{4}.$$

Multiply by A :

$$E_{F=1}^{(0)} = +\frac{1}{4}A, \quad E_{F=0}^{(0)} = -\frac{3}{4}A.$$

Hyperfine splitting $E_{F=1} - E_{F=0} = A = h\nu_{\text{HF}}$ identifies A .

Apply projection theorem within fixed F to the Zeeman terms:

$$\langle F m_F | J_z | F m_F \rangle = \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} m_F.$$

Read off the Landé g_F :

$$g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)} - \frac{m_e}{m_p} g_I \frac{F(F+1) + I(I+1) - J(J+1)}{2F(F+1)}.$$

Combine zero-field and Zeeman:

$$\boxed{E(F, m_F) = A \frac{F(F+1) - I(I+1) - J(J+1)}{2} + g_F \mu_B B m_F.} \quad (9)$$

For ground-state H: $g_{F=1} \simeq +1$, $g_{F=0} = 0$ to leading order in m_e/m_p . Triplet splits linearly with B ; singlet unshifted at first order.

(d) Strong-field eigenenergies

In $|m_J, m_I\rangle$ Zeeman is diagonal. Decompose:

$$\mathbf{I} \cdot \mathbf{J} = I_z J_z + \frac{1}{2}(I_+ J_- + I_- J_+).$$

Drop ladder pieces (off-diagonal in the strong-field basis):

$$\langle m_J, m_I | \mathbf{I} \cdot \mathbf{J} | m_J, m_I \rangle = m_J m_I.$$

Add the Zeeman terms:

$$E(m_J, m_I) = A m_J m_I + g_J \mu_B B m_J - g_I \mu_N B m_I.$$

Insert the four sign combinations:

$$E_1 = E(+\frac{1}{2}, +\frac{1}{2}) = +\frac{1}{4}A + \frac{1}{2}g_J \mu_B B - \frac{1}{2}g_I \mu_N B, \quad (10)$$

$$E_2 = E(+\frac{1}{2}, -\frac{1}{2}) = -\frac{1}{4}A + \frac{1}{2}g_J \mu_B B + \frac{1}{2}g_I \mu_N B, \quad (11)$$

$$E_3 = E(-\frac{1}{2}, +\frac{1}{2}) = -\frac{1}{4}A - \frac{1}{2}g_J \mu_B B - \frac{1}{2}g_I \mu_N B, \quad (12)$$

$$E_4 = E(-\frac{1}{2}, -\frac{1}{2}) = +\frac{1}{4}A - \frac{1}{2}g_J \mu_B B + \frac{1}{2}g_I \mu_N B. \quad (13)$$

Stretched states $|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$ ($|m_F| = 1$) are exact eigenstates at all B ; the $m_F = 0$ pair is given exactly by Breit-Rabi.

(e) Forces and the H maser

High-field upper states E_1, E_2 have $m_J = +1/2$; magnetic moment $\mu_z \simeq -\mu_B$ (electron dominates). Differentiate $E(m_J, m_I)$ w.r.t. B :

$$\frac{\partial E}{\partial B} = g_J \mu_B m_J - g_I \mu_N m_I.$$

Apply $F_z = -(\partial E / \partial z) = -(\partial E / \partial B) \partial_z B_z$:

$$F_z = -g_J \mu_B m_J \frac{\partial B_z}{\partial z} + g_I \mu_N m_I \frac{\partial B_z}{\partial z}.$$

Take leading electronic piece, $g_J m_J = +1$ at $m_J = +1/2$:

$$F_z \simeq -\mu_B \frac{\partial B_z}{\partial z}.$$

Divide by m_p to get acceleration:

$$a \simeq \frac{\mu_B}{m_p} \frac{\partial B_z}{\partial z} = \frac{9.27 \times 10^{-24} \text{ J/T}}{1.67 \times 10^{-27} \text{ kg}} \cdot 20 \text{ T/m}.$$

Compute the ratio:

$$\frac{\mu_B}{m_p} \simeq 5.55 \times 10^3 \text{ m s}^{-2}/(\text{T/m}).$$

Multiply by the gradient:

$$a \simeq 5.55 \times 10^3 \cdot 20 \text{ m s}^{-2}.$$

Collect:

$$\boxed{|a| \simeq 1.1 \times 10^5 \text{ m s}^{-2} \text{ for both upper states.}} \quad (14)$$

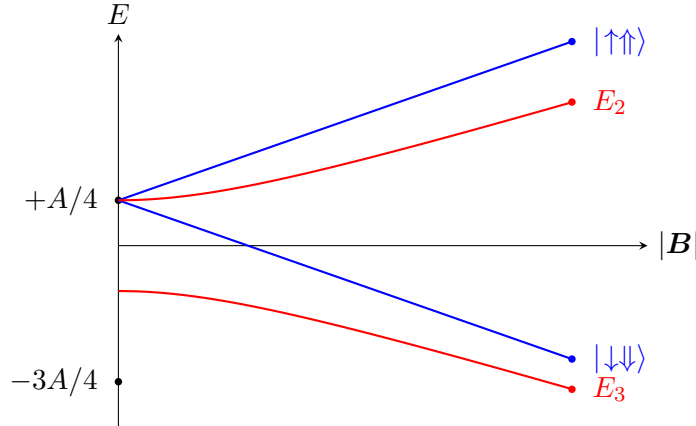
Force difference between E_1 ($m_I = +1/2$) and E_2 ($m_I = -1/2$):

$$\Delta F = F_z(E_1) - F_z(E_2) = g_I \mu_N \frac{\partial B_z}{\partial z}.$$

Divide by the mean $\langle F \rangle \simeq g_J \mu_B m_J \partial_z B_z$:

$$\frac{\Delta F}{\langle F \rangle} \simeq \frac{g_I \mu_N}{g_J \mu_B m_J} \sim \frac{m_e}{m_p} \sim \frac{1}{1836} \sim 5 \times 10^{-4}.$$

Breit–Rabi connection. Stretched states $|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$ correspond to $|F=1, m_F=\pm 1\rangle$. The $m_F=0$ pair: E_2 at high field connects to $|F=1, m_F=0\rangle$; E_3 connects to $|F=0, m_F=0\rangle$.



Hydrogen maser. The Stern–Gerlach gradient deflects low-field-seeking states E_1, E_2 ($m_J = +1/2$) into a PTFE-coated storage bulb. Inside a low-field microwave cavity tuned to $\nu_{\text{hf}} = 1420 \text{ MHz}$, only the upper $F=1$ levels are populated, the $|F=0, m_F=0\rangle$ ground-hyperfine state is empty, and the $\Delta F=1, \Delta m_F=0$ M1 transition is driven by stimulated emission. Wall-collision lifetimes of $\sim \text{s}$ give an extremely narrow Lorentzian and a precision frequency standard.

Question 3 — Three- vs. four-level lasers and the Yb fibre

Problem. (a) Threshold contrast between three- and four-level systems. (b) Derive the gain coefficient from the Einstein relations. (c)–(g) Yb-doped fibre: $r = 2 \mu\text{m}$, $L = 0.5 \text{ m}$, $\sigma_{\text{abs}} = 6 \times 10^{-25} \text{ m}^2$, $\sigma_{\text{em}} = 2.5 \times 10^{-25} \text{ m}^2$, 86% pump absorption at 910 nm , lower-band sublevels at $E_{1A}/hc = 60\,000 \text{ m}^{-1}$, $E_{1B}/hc = 106\,000 \text{ m}^{-1}$, upper level at $1\,026\,000 \text{ m}^{-1}$.

(a) Threshold for three- vs. four-level lasers

Inversion $N^* = N_2 - (g_2/g_1)N_1$ must overcome cavity loss.

Four-level: $|1\rangle$ above the ground state, thermally empty ($N_1 \sim e^{-\Delta E/k_B T} \rightarrow 0$); inversion needs an infinitesimal fraction of ground population pumped.

Three-level: $|1\rangle$ is the ground state; need $N_2 > N_1$, i.e. $> \frac{1}{2}$ of all ions pumped. Threshold pump rate $\sim$ spontaneous-decay rate $\times N/2$, orders of magnitude larger.

(b) Gain coefficient

A monochromatic beam of intensity I has energy density I/c concentrated near ω . Stimulated rate per atom on a homogeneously broadened line:

$$W_{\text{stim}} = B_{21} \rho_\omega g(\omega - \omega_0) = B_{21} \frac{I}{c} g(\omega - \omega_0),$$

with $\int g d\omega = 1$.

Net rate of photons per volume into the beam:

$$\frac{1}{\hbar\omega_0} \frac{dI}{dz} = (B_{21}N_2 - B_{12}N_1) \frac{I}{c} g(\omega - \omega_0).$$

Multiply by $\hbar\omega_0$:

$$\frac{dI}{dz} = \hbar\omega_0 (B_{21}N_2 - B_{12}N_1) \frac{I}{c} g(\omega - \omega_0).$$

Assumptions: (i) plane wave; (ii) homogeneous broadening; (iii) populations not depleted; (iv) thermal photons negligible at optical frequency.

Define $\gamma(\omega)$ by $dI/dz = \gamma I$:

$$\gamma(\omega) = \frac{\hbar\omega_0 g(\omega - \omega_0)}{c} (B_{21}N_2 - B_{12}N_1).$$

Apply $g_1 B_{12} = g_2 B_{21}$:

$$\gamma(\omega) = \frac{\hbar\omega_0 g(\omega - \omega_0)}{c} B_{21} \left[N_2 - \frac{g_2}{g_1} N_1 \right].$$

Eliminate B_{21} via $A_{21} = \hbar\omega_0^3 B_{21}/(\pi^2 c^3)$, i.e. $B_{21} = \pi^2 c^3 A_{21}/(\hbar\omega_0^3)$:

$$\gamma(\omega) = \frac{\hbar\omega_0 g(\omega - \omega_0)}{c} \frac{\pi^2 c^3 A_{21}}{\hbar\omega_0^3} \left[N_2 - \frac{g_2}{g_1} N_1 \right].$$

Cancel $\hbar\omega_0$ and one factor of c :

$$\boxed{\gamma(\omega) = \frac{\pi^2 c^2 A_{21}}{\omega_0^2} g(\omega - \omega_0) \left[N_2 - \frac{g_2}{g_1} N_1 \right] = \sigma_{\text{em}}(\omega) N^*} \quad (15)$$

(c) Yb ion density

Beer absorption with 86% absorbed gives $I(L)/I(0) = 0.14$:

$$e^{-\sigma_{\text{abs}} N L} = 0.14.$$

Take the log:

$$\sigma_{\text{abs}} N L = -\ln 0.14 = 1.97.$$

Solve for N :

$$N = \frac{1.97}{\sigma_{\text{abs}} L} = \frac{1.97}{6 \times 10^{-25} \text{ m}^2 \cdot 0.5 \text{ m}}.$$

Evaluate the denominator:

$$N = \frac{1.97}{3 \times 10^{-25} \text{ m}^3}.$$

Compute the quotient:

$$\boxed{N \simeq 6.6 \times 10^{24} \text{ m}^{-3}}. \quad (16)$$

(d) Lower-band thermal populations at 300 K

Three lower-band sublevels $\{0, 1A, 1B\}$ in thermal equilibrium, all degeneracies 1.

Convert $k_B T$ to wavenumber with $k_B T = 4.14 \times 10^{-21} \text{ J}$ and $hc = 1.99 \times 10^{-25} \text{ J m}$:

$$\frac{k_B T}{hc} = \frac{4.14 \times 10^{-21}}{1.99 \times 10^{-25}} \simeq 2.084 \times 10^4 \text{ m}^{-1}.$$

Form dimensionless exponents:

$$\begin{aligned} \frac{E_{1A}}{k_B T} &= \frac{60\,000}{2.084 \times 10^4} \simeq 2.88. \\ \frac{E_{1B}}{k_B T} &= \frac{106\,000}{2.084 \times 10^4} \simeq 5.09. \end{aligned}$$

Form Boltzmann factors:

$$e^{-2.88} \simeq 0.0561, \quad e^{-5.09} \simeq 0.00614.$$

Partition sum over $\{0, 1A, 1B\}$:

$$Z_{\text{low}} = 1 + 0.0561 + 0.00614 \simeq 1.062.$$

Normalise to get fractional populations $f_\alpha = e^{-E_\alpha/k_B T} / Z_{\text{low}}$:

$$\begin{aligned} f_{1A} &= \frac{0.0561}{1.062} \simeq 5.3 \times 10^{-2}. \\ f_{1B} &= \frac{0.00614}{1.062} \simeq 5.8 \times 10^{-3}. \end{aligned}$$

$$\boxed{f_{1A} \simeq 5.3 \times 10^{-2}, \quad f_{1B} \simeq 5.8 \times 10^{-3}}. \quad (17)$$

(e) Threshold pump energies

Threshold condition $N_{2,\text{th}} = N_{1\alpha,\text{th}}$ with $g_2/g_{1\alpha} = 1$. The lower-band ions thermalise, so $N_{1\alpha} = f_{1\alpha}(N - N_2) \simeq f_{1\alpha}N$ for $f_{1\alpha} \ll 1$:

$$N_{2,\text{th}} \simeq f_{1\alpha}N.$$

Compute pump-photon energy at 910 nm:

$$E_{\text{pump}} = \frac{hc}{\lambda_{\text{pump}}} = \frac{6.626 \times 10^{-34} \cdot 3 \times 10^8}{910 \times 10^{-9}}.$$

Evaluate:

$$E_{\text{pump}} \simeq 2.18 \times 10^{-19} \text{ J} \simeq 1.36 \text{ eV}.$$

Core volume:

$$V = \pi r^2 L = \pi(2 \times 10^{-6} \text{ m})^2 \cdot 0.5 \text{ m}.$$

Evaluate:

$$V \simeq 6.28 \times 10^{-12} \text{ m}^3.$$

Total ions in the core:

$$NV \simeq 6.6 \times 10^{24} \text{ m}^{-3} \cdot 6.28 \times 10^{-12} \text{ m}^3 \simeq 4.14 \times 10^{13}.$$

Threshold pump energy $E_{\alpha, \text{th}} = N_{2, \text{th}} V E_{\text{pump}} = f_{1\alpha} N V E_{\text{pump}}$:

$$E_{A, \text{th}} = 5.3 \times 10^{-2} \cdot 4.14 \times 10^{13} \cdot 2.18 \times 10^{-19} \text{ J}.$$

Multiply factors for A :

$$E_{A, \text{th}} \simeq 4.78 \times 10^{-7} \text{ J}.$$

Compute B :

$$E_{B, \text{th}} = 5.8 \times 10^{-3} \cdot 4.14 \times 10^{13} \cdot 2.18 \times 10^{-19} \text{ J}.$$

Multiply factors for B :

$$E_{B, \text{th}} \simeq 5.23 \times 10^{-8} \text{ J}.$$

$$\boxed{E_{A, \text{th}} \simeq 0.48 \text{ }\mu\text{J}, \quad E_{B, \text{th}} \simeq 52 \text{ nJ.}}$$

(18)

(f) Round-trip gain at $E = 5E_{B, \text{th}}$

Pump $5 \times$ threshold for B excites $N_2 = 5f_{1B}N$:

$$N_2 = 5 \cdot 5.8 \times 10^{-3} \cdot 6.6 \times 10^{24} \text{ m}^{-3}.$$

Multiply factors:

$$N_2 \simeq 1.9 \times 10^{23} \text{ m}^{-3}.$$

Remaining ions in lower band:

$$N - N_2 = 6.6 \times 10^{24} - 0.19 \times 10^{24} \simeq 6.41 \times 10^{24} \text{ m}^{-3}.$$

Thermal lower-band populations:

$$N_{1A} = f_{1A}(N - N_2) \simeq 5.3 \times 10^{-2} \cdot 6.41 \times 10^{24} \text{ m}^{-3} \simeq 3.4 \times 10^{23} \text{ m}^{-3}.$$

$$N_{1B} = f_{1B}(N - N_2) \simeq 5.8 \times 10^{-3} \cdot 6.41 \times 10^{24} \text{ m}^{-3} \simeq 3.7 \times 10^{22} \text{ m}^{-3}.$$

Inversions ($g_2/g_{1\alpha} = 1$):

$$N_A^* = N_2 - N_{1A} = 1.9 \times 10^{23} \text{ m}^{-3} - 3.4 \times 10^{23} \text{ m}^{-3} \simeq -1.5 \times 10^{23} \text{ m}^{-3} < 0.$$

$$N_B^* = N_2 - N_{1B} = 1.9 \times 10^{23} \text{ m}^{-3} - 3.7 \times 10^{22} \text{ m}^{-3} \simeq +1.5 \times 10^{23} \text{ m}^{-3}.$$

Round-trip gain on $2 \rightarrow 1B$:

$$G_{\text{rt}} = \exp(2\sigma_{\text{em}} N_B^* L) = \exp(2 \cdot 2.5 \times 10^{-25} \text{ m}^2 \cdot 1.5 \times 10^{23} \text{ m}^{-3} \cdot 0.5 \text{ m}).$$

Multiply out the exponent:

$$2\sigma_{\text{em}} N_B^* L = 2 \cdot 0.5 \cdot 2.5 \cdot 1.5 \times 10^{-2} = 3.75 \times 10^{-2}.$$

Exponentiate:

$$\boxed{G_{\text{rt}} \simeq e^{0.0375} \simeq 1.04 (\sim 4\%); \text{ no gain on } 2 \rightarrow 1A.}$$

(19)

(g) Maximum efficiency and losses

Quantum-defect efficiency is the ratio of laser to pump photon energy:

$$\eta_{\text{QD}} = \frac{E_{\text{laser}}}{E_{\text{pump}}} = \frac{\lambda_{\text{pump}}}{\lambda_{\text{laser}}}.$$

Compute laser wavenumber $2 \rightarrow 1B$:

$$\frac{E_{\text{laser}}^B}{hc} = (1\,026\,000 - 106\,000) \text{ m}^{-1} = 920\,000 \text{ m}^{-1}.$$

Compute pump wavenumber at $\lambda_{\text{pump}} = 910 \text{ nm}$:

$$\frac{E_{\text{pump}}}{hc} = \frac{1}{\lambda_{\text{pump}}} = \frac{1}{910 \times 10^{-9} \text{ m}} \simeq 1\,099\,000 \text{ m}^{-1}.$$

Form the ratio:

$$\eta_{\text{QD}}^B = \frac{920\,000}{1\,099\,000} \simeq 0.837.$$

Compute laser wavenumber $2 \rightarrow 1A$:

$$\frac{E_{\text{laser}}^A}{hc} = (1\,026\,000 - 60\,000) \text{ m}^{-1} = 966\,000 \text{ m}^{-1}.$$

Form the ratio:

$$\eta_{\text{QD}}^A = \frac{966\,000}{1\,099\,000} \simeq 0.879.$$

Multiply by $\eta_{\text{diode}} = 0.60$ and $\eta_{\text{abs}} = 0.86$ for B :

$$\eta_{\text{max}}^B = 0.60 \cdot 0.86 \cdot 0.837 \simeq 0.43.$$

Repeat for A :

$$\eta_{\text{max}}^A = 0.60 \cdot 0.86 \cdot 0.879 \simeq 0.45.$$

$$\boxed{\eta_{\text{max}} \simeq 43\text{--}45\%}. \quad (20)$$

Practical losses. Ground-state absorption on the laser line (especially $2 \rightarrow 1B$, quasi-three-level); excited-state absorption from level 2; non-radiative quenching by impurities or Yb clusters; cavity scattering; sub-optimal output coupling; pump-heating broadening of the emission line.

Question 4 — Radiative transfer and lithium absorption

Problem. Recast $dI_\nu/dz = -\alpha_\nu I_\nu + j_\nu$ in optical-depth form, solve for constant S_ν , and use measured ratios $I_A : I_B : I_C = 0.85 : 1 : 0.15$ to extract τ_ν and $S_\nu/I_\nu(0)$. Show $\sigma_{\text{abs}}^{\text{res}} = \lambda_0^2/(2\pi)$ for a lifetime-broadened line, evaluate for $\text{Li } 2s \rightarrow 2p$ at 671 nm, give the column density for $\tau \sim 1$ in a 1 cm cell, and discuss when natural-width lines are observed; list the source requirements for the $\text{Li } 2s \rightarrow np$ Rydberg series.

(a) Optical-depth form and formal solution

Define:

$$d\tau_\nu = \alpha_\nu dz, \quad S_\nu = \frac{j_\nu}{\alpha_\nu}.$$

Divide transfer equation by α_ν :

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu.$$

Multiply by integrating factor e^{τ_ν} :

$$\frac{d}{d\tau_\nu}(I_\nu e^{\tau_\nu}) = S_\nu e^{\tau_\nu}.$$

Integrate from 0 to τ_ν at constant S_ν :

$$I_\nu(\tau_\nu)e^{\tau_\nu} - I_\nu(0) = S_\nu(e^{\tau_\nu} - 1).$$

Solve for I_ν :

$$\boxed{I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + S_\nu(1 - e^{-\tau_\nu}).} \quad (21)$$

Beer's law in the limit $\tau_\nu \rightarrow 0$; black-body S_ν in the limit $\tau_\nu \rightarrow \infty$.

(b) Inversion of the flame measurement

Identify the three measurements with (21):

$$I_A = I_\nu(0)e^{-\tau_\nu} + S_\nu(1 - e^{-\tau_\nu}), \quad I_B = I_\nu(0), \quad I_C = S_\nu(1 - e^{-\tau_\nu}).$$

Subtract I_C from I_A :

$$I_A - I_C = I_B e^{-\tau_\nu}.$$

Divide by I_B , set $X = I_A/I_B$, $Y = I_C/I_B$:

$$e^{-\tau_\nu} = X - Y.$$

Insert numerical values:

$$e^{-\tau_\nu} = 0.85 - 0.15 = 0.70.$$

Take log:

$$\boxed{\tau_\nu = -\ln 0.70 \simeq 0.357.} \quad (22)$$

Read $S_\nu/I_\nu(0)$ from I_C/I_B :

$$\frac{S_\nu}{I_\nu(0)} = \frac{Y}{1 - e^{-\tau_\nu}} = \frac{Y}{1 - (X - Y)}.$$

Insert numerical values:

$$\frac{S_\nu}{I_\nu(0)} = \frac{0.15}{1 - 0.70} = \frac{0.15}{0.30}.$$

Evaluate:

$$\boxed{\frac{S_\nu}{I_\nu(0)} = 0.50.} \quad (23)$$

Optically thin flame; carbon emits with half the incoming black-body brightness at this frequency.

(c) Resonant cross section and the Li $2s \rightarrow 2p$ line

Natural-width Lorentzian normalised in ω , $\int g_L d\omega = 1$, with $\Gamma = A_{21}$:

$$g_L(\omega - \omega_0) = \frac{1}{2\pi} \frac{\Gamma}{(\omega - \omega_0)^2 + (\Gamma/2)^2}.$$

Set $\omega = \omega_0$:

$$g_L(0) = \frac{\Gamma/(2\pi)}{(\Gamma/2)^2} = \frac{\Gamma/(2\pi)}{\Gamma^2/4}.$$

Simplify:

$$g_L(0) = \frac{2}{\pi\Gamma} = \frac{2}{\pi A_{21}}.$$

Insert in $\sigma_{\text{abs}} = (g_2/g_1)(\pi^2 c^2 A_{21}/\omega_0^2) g_L$, take $g_2/g_1 = 1$:

$$\sigma_{\text{abs}}^{\text{res}} = \frac{\pi^2 c^2 A_{21}}{\omega_0^2} \cdot \frac{2}{\pi A_{21}}.$$

Cancel A_{21} and one π :

$$\sigma_{\text{abs}}^{\text{res}} = \frac{2\pi c^2}{\omega_0^2}.$$

Use $\omega_0 = 2\pi c/\lambda_0$, so $\omega_0^2 = 4\pi^2 c^2/\lambda_0^2$:

$$\sigma_{\text{abs}}^{\text{res}} = \frac{2\pi c^2 \lambda_0^2}{4\pi^2 c^2}.$$

Cancel:

$$\boxed{\sigma_{\text{abs}}^{\text{res}} = \frac{\lambda_0^2}{2\pi} \text{ (independent of } A_{21}\text{)}} \quad (24)$$

For Li, $\lambda_0 = 671 \text{ nm}$:

$$\sigma_{\text{abs}}^{\text{res}} = \frac{(671 \times 10^{-9} \text{ m})^2}{2\pi} = \frac{4.50 \times 10^{-13} \text{ m}^2}{2\pi}.$$

Divide:

$$\boxed{\sigma_{\text{abs}}^{\text{res}} \simeq 7.2 \times 10^{-14} \text{ m}^2}. \quad (25)$$

Column density for $\tau \sim 1$ at $L = 1 \text{ cm}$:

$$N \sim \frac{1}{\sigma_{\text{abs}}^{\text{res}} L} = \frac{1}{7.2 \times 10^{-14} \text{ m}^2 \cdot 10^{-2} \text{ m}}.$$

Compute the denominator:

$$N \sim \frac{1}{7.2 \times 10^{-16} \text{ m}^3}.$$

Evaluate:

$$\boxed{N_{\text{Li}} \sim 1.4 \times 10^{15} \text{ m}^{-3} (\sim 10^9 \text{ cm}^{-3})}. \quad (26)$$

When are pure lifetime-broadened lines seen? Natural width must dominate Doppler ($\propto \sqrt{T/M}$) and pressure ($\propto N\sigma_{\text{coll}}\bar{v}$) widths: very low temperature (laser cooling, transverse atomic beams with $\mathbf{k} \cdot \mathbf{v} \rightarrow 0$), very low pressure ($\tau_{\text{coll}} \gg \tau_{\text{rad}}$), and absence of inhomogeneous mechanisms (Stark, transit-time). Examples: forbidden lines via saturated absorption, Ramsey spectroscopy, single-ion traps.

(d) Source for the $2s \rightarrow np$ Li series

Li ground state $1s^2 2s$, ionisation $5.39 \text{ eV} \equiv 43\,484 \text{ cm}^{-1}$, series limit $\lambda_{\text{lim}} \simeq 230 \text{ nm}$.

Required source properties:

(i) *Continuum coverage* from $\sim 671 \text{ nm}$ down to $\sim 230 \text{ nm}$: D_2 arc, Xe arc, tungsten + UV continuum, or synchrotron radiation, so absorption appears as dips on a smooth background.

(ii) *High UV brightness*: peak cross section $\propto 1/n^{*5}$ for high Rydberg states; need adequate UV signal-to-noise.

(iii) *Featureless/stable spectrum*: any source structure mimics absorption; alternative is a swept tunable laser of sufficient range.

(iv) *Resolving power* $\Delta\nu/\nu \lesssim 10^{-6}$ to resolve Doppler width at $T \sim 700 \text{ K}$: Echelle, Fourier-transform, or two-photon Doppler-free laser spectroscopy.

(v) *Long-term intensity stability*: faint high- n lines need steady I/I_0 ratios over long acquisition.

In short: continuum or broadly tunable laser covering 230–700 nm, high UV brightness, excellent stability, recorded at high resolving power against a wavelength reference.